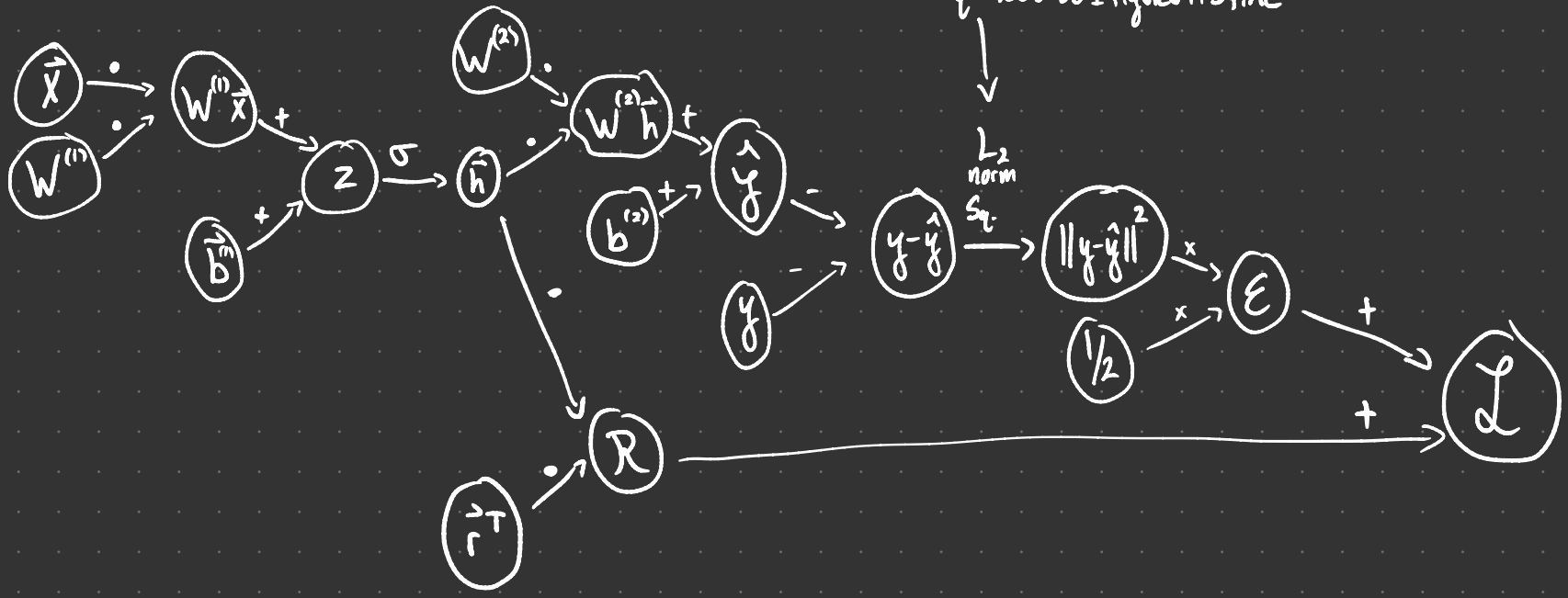
AI 360 HW1

jadenvm
94916335

Q1:

a)



b) need $\nabla_{W^1}\mathcal{L}$, $\nabla_{b^1}\mathcal{L}$, $\nabla_{W^2}\mathcal{L}$, $\nabla_{b^2}\mathcal{L}$ for backprop

$$\frac{\partial \mathcal{L}}{\partial x} = \frac{\partial \mathcal{L}}{\partial \mathcal{E}} + \frac{\partial \mathcal{L}}{\partial R} = \frac{\partial \mathcal{L}}{\partial \mathcal{E}}\frac{\partial \mathcal{E}}{\partial x} + \frac{\partial \mathcal{L}}{\partial R}\frac{\partial R}{\partial x}$$

$$\frac{\partial \mathcal{E}}{\partial x} = \frac{\partial \mathcal{E}}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial x} = (y-\hat{y})\frac{\partial \hat{y}}{\partial x}$$

$$\frac{\partial \hat{y}}{\partial x} = \frac{\partial \hat{y}}{\partial h}\frac{\partial h}{\partial x} = W^{(2)T}\frac{\partial h}{\partial x}$$

$$\frac{\partial h}{\partial x} = \frac{\partial h}{\partial z}\frac{\partial z}{\partial x} = z\odot(1-z)\odot\frac{\partial z}{\partial x}$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial x} = W^{(1)}$$

$$\frac{\partial R}{\partial x} = \frac{\partial R}{\partial h}\frac{\partial h}{\partial x} = r^T\mathbb{1}\frac{\partial h}{\partial x}$$

$$= r^T\mathbb{1}z(1-z)W^{(1)T}$$

$$\frac{\partial \mathcal{L}}{\partial x} = W^{(1)T}\left(\left(\left(W^{(2)T}(\hat{y}-y)+r\right)\odot h\odot(1-h)\right)\right.$$

Q2: See other file